

PREFACE FOR FACULTY

The Cooley-Tukey Radix-2 Decimation-in-Time (DIT) Fast Fourier Transform is one of the foundational algorithms in digital signal processing. It exploits the divide-and-conquer strategy by recursively decomposing an N -point DFT into smaller sub-DFTs of even- and odd-indexed sequences.

Pedagogical Strategy:

- 1. Start with the Elementary 2-Point DFT ($N = 2$):** Build the fundamental butterfly processing element.
 - 2. Scale to the 4-Point DIT-FFT ($N = 4$):** Show how two 2-point DFTs combine in 2 stages with twiddle factors $W_4^0 = 1$ and $W_4^1 = -j$.
 - 3. Scale to the 8-Point DIT-FFT ($N = 8$):** Show how two 4-point DFTs (and four 2-point DFTs) combine in 3 stages with twiddle factors $W_8^0, W_8^1, W_8^2, W_8^3$.
 - 4. Master Bit-Reversal Indexing:** Binary transposition rule $n \leftrightarrow n_{\text{rev}}$.
 - 5. Emphasize In-Place Memory Computation:** Registers are overwritten stage-by-stage without allocating external memory buffers.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- 1. Derive** and compute the 2-point, 4-point, and 8-point DIT-FFT butterfly equations.
 - 2. Construct** complete signal flow graphs (SFG) for $N = 2, 4, 8$ with accurate twiddle factors and branch signs.
 - 3. Compute** bit-reversed index mappings for $N = 2, 4, 8, 16$.
 - 4. Evaluate** intermediate and final butterfly outputs for given numerical sequences step-by-step.
 - 5. Analyze** arithmetic complexity savings ($\frac{N}{2}\log_2 N$ vs N^2).
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2. MATHEMATICAL FOUNDATIONS & SIGNAL FLOW GRAPHS

2.1 General Radix-2 DIT Decomposition

Let $x[n]$ be an N -point sequence ($N = 2^M$). Splitting $x[n]$ into even and odd samples:

$$X[k] = \sum_{r=0}^{N/2-1} x[2r]W_{N/2}^{rk} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1]W_{N/2}^{rk} = X_e[k] + W_N^k X_o[k]$$

Using the twiddle symmetry $W_N^{k+N/2} = -W_N^k$:

$$X[k] = X_e[k] + W_N^k X_o[k], \quad 0 \leq k < N/2$$

$$X[k + N/2] = X_e[k] - W_N^k X_o[k], \quad 0 \leq k < N/2$$

2.2 The 2-Point FFT ($N = 2$, 1 Stage, 1 Butterfly)

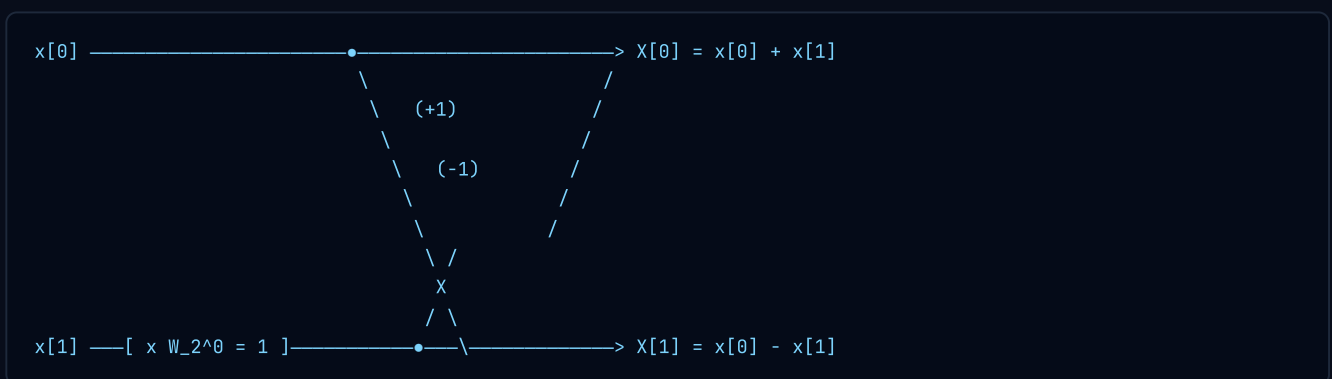
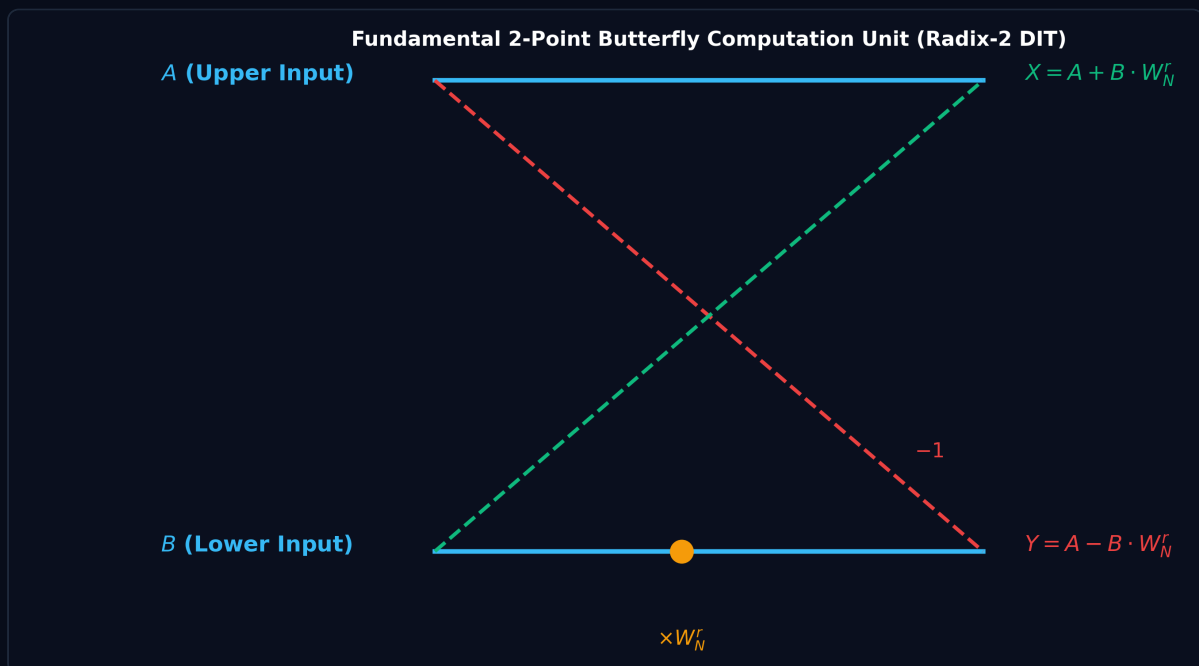
For $N = 2$, $M = \log_2 2 = 1$ stage. The twiddle factor is $W_2^0 = e^{-j0} = 1$.

Mathematical Equations:

$$X[0] = x[0] + W_2^0 x[1] = x[0] + x[1]$$

$$X[1] = x[0] - W_2^0 x[1] = x[0] - x[1]$$

2-Point Signal Flow Graph (The Basic Butterfly Unit):



- **Complexity:** 1 addition, 1 subtraction, 0 non-trivial multiplications.

2.3 The 4-Point DIT-FFT ($N = 4$, 2 Stages, 4 Butterflies)

For $N = 4$, $M = \log_2 4 = 2$ stages.

- **Twiddle Factors:**

- $W_4^0 = e^{-j0} = 1$
- $W_4^1 = e^{-j\frac{2\pi}{4}} = e^{-j\pi/2} = -j$
- **Bit-Reversal Permutation (2-bit):**
 - $n = 0, (00)_2 \rightarrow 0, (00)_2 \Rightarrow x[0]$
 - $n = 1, (01)_2 \rightarrow 2, (10)_2 \Rightarrow x[2]$
 - $n = 2, (10)_2 \rightarrow 1, (01)_2 \Rightarrow x[1]$
 - $n = 3, (11)_2 \rightarrow 3, (11)_2 \Rightarrow x[3]$ Input array ordering: $x[0], x[2], x[1], x[3]$.

Stage-by-Stage Equations:

- **Stage 1 (Two 2-point Sub-DFTs, Twiddle $W_4^0 = 1$):**

- Upper 2-pt DFT:

$$v_1[0] = x[0] + W_4^0 x[2] = x[0] + x[2]$$

$$v_1[1] = x[0] - W_4^0 x[2] = x[0] - x[2]$$

- Lower 2-pt DFT:

$$v_1[2] = x[1] + W_4^0 x[3] = x[1] + x[3]$$

$$v_1[3] = x[1] - W_4^0 x[3] = x[1] - x[3]$$

- **Stage 2 (One 4-point DFT, Twiddles $W_4^0 = 1, W_4^1 = -j$):**

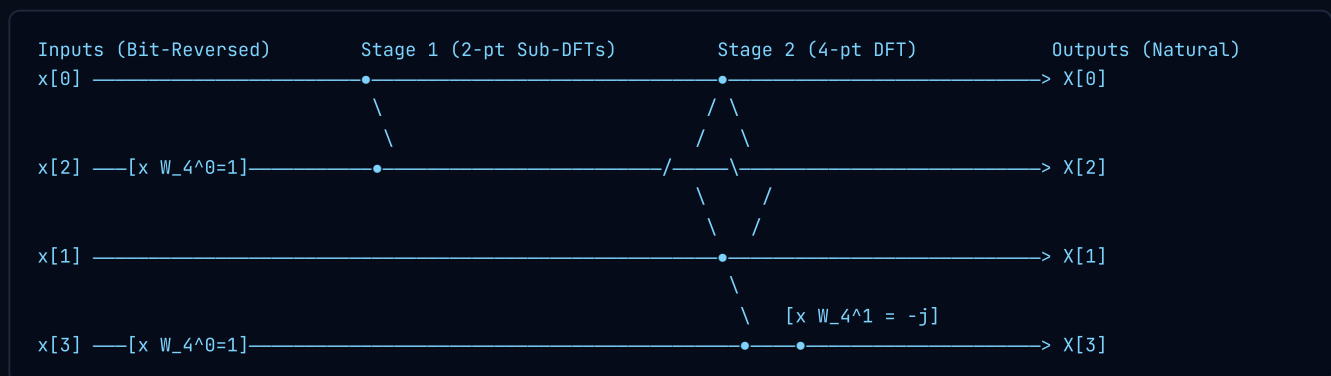
$$X[0] = v_1[0] + W_4^0 v_1[2] = v_1[0] + v_1[2]$$

$$X[1] = v_1[1] + W_4^1 v_1[3] = v_1[1] - jv_1[3]$$

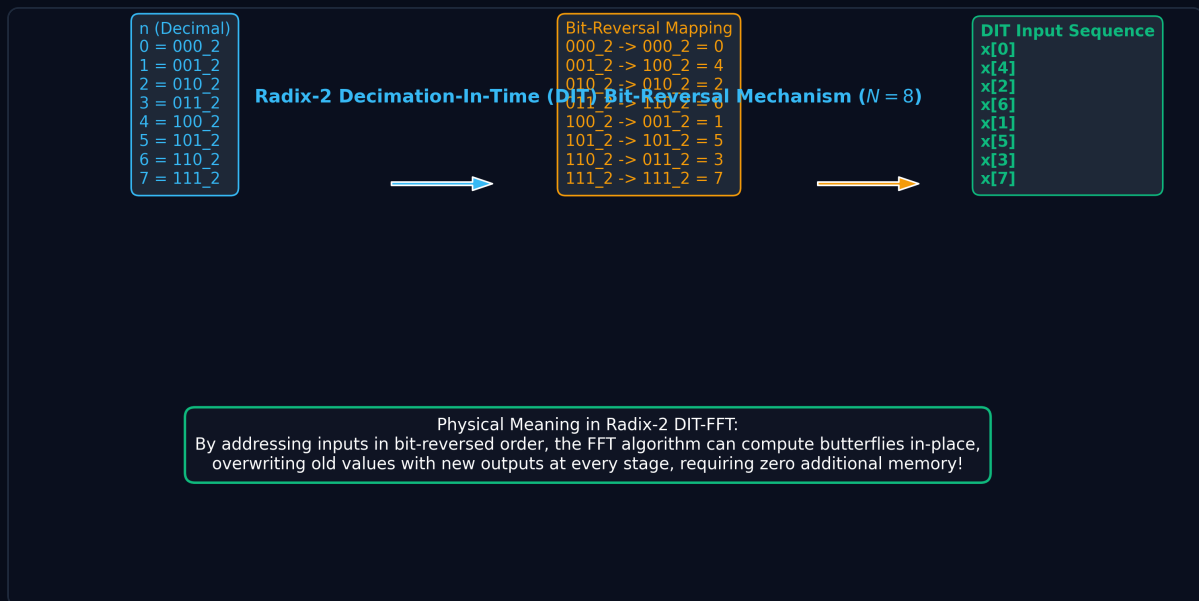
$$X[2] = v_1[0] - W_4^0 v_1[2] = v_1[0] - v_1[2]$$

$$X[3] = v_1[1] - W_4^1 v_1[3] = v_1[1] + jv_1[3]$$

Complete 4-Point DIT-FFT Signal Flow Graph:

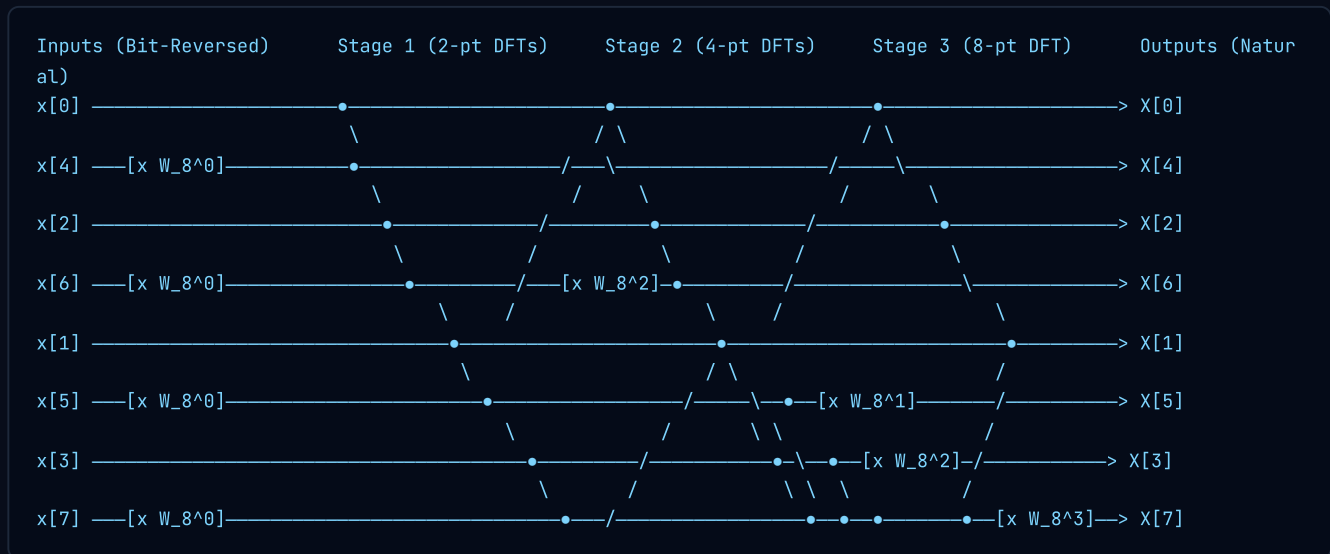


2.4 The 8-Point DIT-FFT ($N = 8$, 3 Stages, 12 Butterflies)



- **Bit-Reversal Permutation (3-bit):** $x_{\text{rev}} = [x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]]$.
- **Twiddle Factors:**
 - $W_8^0 = 1$
 - $W_8^1 = e^{-j\pi/4} = \frac{\sqrt{2}}{2}(1 - j) \approx 0.7071 - 0.7071j$
 - $W_8^2 = e^{-j\pi/2} = -j$
 - $W_8^3 = e^{-j3\pi/4} = -\frac{\sqrt{2}}{2}(1 + j) \approx -0.7071 - 0.7071j$

Complete 8-Point DIT-FFT Signal Flow Graph:



3. WORKED NUMERICAL EXAMPLES

Example 10.1: 2-Point FFT Computation

Problem: Compute the 2-point DFT of $x[n] = \underset{\uparrow}{3}, 1$ using the butterfly equations.

Solution:

- $X[0] = x[0] + x[1] = 3 + 1 = 4$
- $X[1] = x[0] - x[1] = 3 - 1 = 2$
- **Result:** $X[k] = \underset{\uparrow}{4}, 2.$

Example 10.2: 4-Point DIT-FFT Step-by-Step

Problem: Compute the 4-point DFT of $x[n] = \underset{\uparrow}{1}, 2, 3, 4$ using the Radix-2 DIT-FFT algorithm.

Solution:

1. Bit-Reversed Input Array:

$$x_{\text{rev}} = [x[0], x[2], x[1], x[3]] = [1, 3, 2, 4]$$

2. Stage 1 (Two 2-Point Butterflies with $W_4^0 = 1$):

- Upper pair $(x[0], x[2])$:

$$v_1[0] = 1 + 3 = 4$$

$$v_1[1] = 1 - 3 = -2$$

- Lower pair $(x[1], x[3])$:

$$v_1[2] = 2 + 4 = 6$$

$$v_1[3] = 2 - 4 = -2$$

Stage 1 output: $v_1 = [4, -2, 6, -2]$.

3. Stage 2 (One 4-Point Butterfly with $W_4^0 = 1, W_4^1 = -j$):

- $X[0] = v_1[0] + W_4^0 v_1[2] = 4 + 1(6) = 10$
- $X[1] = v_1[1] + W_4^1 v_1[3] = -2 + (-j)(-2) = -2 + 2j$
- $X[2] = v_1[0] - W_4^0 v_1[2] = 4 - 1(6) = -2$
- $X[3] = v_1[1] - W_4^1 v_1[3] = -2 - (-j)(-2) = -2 - 2j$

• Final Result:

$$X[k] = \underset{\uparrow}{10}, -2 + 2j, -2, -2 - 2j$$

Example 10.3: 8-Point DIT-FFT Execution

Problem: Compute the 8-point DFT of $x[n] = 1, 2, 3, 4, 4, 3, 2, 1$ using Radix-2 DIT-FFT.

Solution:

1. Input Bit-Reversal: $x_{\text{rev}} = [x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]] = [1, 4, 3, 2, 2, 3, 4, 1]$.

2. Stage 1 (2-Point Butterflies, $W_8^0 = 1$): $v_1 = [5, -3, 5, 1, 5, -1, 5, 3]$.

3. Stage 2 (4-Point Butterflies, $W_8^0 = 1, W_8^2 = -j$): $v_2 = [10, -3 - j, 0, -3 + j, 10, -1 - 3j, 0, -1 + 3j]$.

4. Stage 3 (8-Point Butterflies, $W_8^0, W_8^1, W_8^2, W_8^3$):

- $X[0] = 10 + 10 = 20$
- $X[1] = (-3 - j) + (0.7071 - 0.7071j)(-1 - 3j) = -5.8284 - 2.4142j$
- $X[2] = 0 + (-j)(0) = 0$

- $X[3] = (-3 + j) + (-0.7071 - 0.7071j)(-1 + 3j) = -\mathbf{0.1716} - \mathbf{0.4142j}$
- $X[4] = 10 - 10 = \mathbf{0}$
- $X[5] = X[3]^* = -\mathbf{0.1716} + \mathbf{0.4142j}$
- $X[6] = X[2]^* = \mathbf{0}$
- $X[7] = X[1]^* = -\mathbf{5.8284} + \mathbf{2.4142j}$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Draw the 2-point butterfly computation block and explain how it is used to construct 4-point and 8-point DIT-FFT networks. *(5 Marks)* **(b)** Given $x[n] = \underset{\uparrow}{1}, 2, 3, 4$, compute its 4-point DFT using the Radix-2 DIT-FFT algorithm. Show all intermediate stage values and draw the labeled signal flow graph. *(10 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - 2-point butterfly diagram with equations $A \pm W_N^r B$ *(3 Marks)*
 - Explanation of hierarchical sub-DFT combining ($N/2 \rightarrow N$) *(2 Marks)*
- **Part (b):**
 - Bit-reversal input ordering 1, 3, 2, 4 *(2 Marks)*
 - Stage 1 calculations: $v_1 = [4, -2, 6, -2]$ *(3 Marks)*
 - Stage 2 calculations with $W_4^0 = 1$ and $W_4^1 = -j$:
 $X[0] = 10, X[1] = -2 + 2j, X[2] = -2, X[3] = -2 - 2j$ *(3 Marks)*
 - Labeled 4-point Signal Flow Graph *(2 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# 2-Point FFT
x2 = np.array([3, 1])
print("2-Point FFT:", np.fft.fft(x2))

# 4-Point FFT
x4 = np.array([1, 2, 3, 4])
print("4-Point FFT:", np.fft.fft(x4))

# 8-Point FFT
x8 = np.array([1, 2, 3, 4, 4, 3, 2, 1])
print("8-Point FFT:", np.round(np.fft.fft(x8), 4))
```